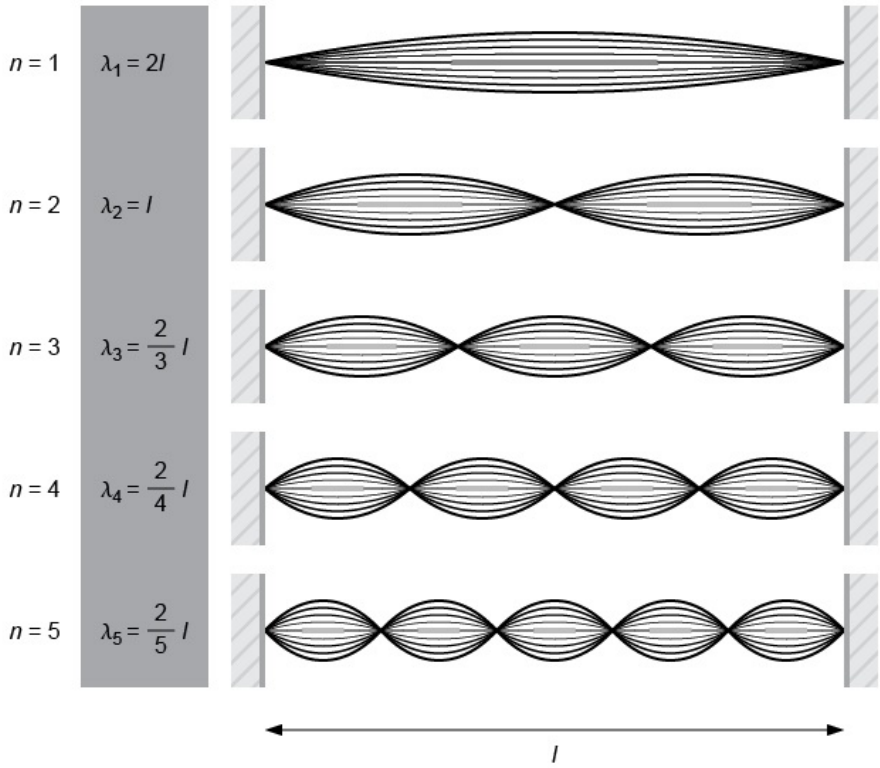
普通物理實驗結報



繩波

第三組

組員:	貢獻度
吳	37.5%
卓	37.5%
程	25 %

1 實驗目的

To learn and observe how different parameters and conditions affect a standing wave's characteristic.

2 實驗原理

Standing waves are waves that don't appear to be moving through the medium, but *oscillate* in place. They form when two identically same waves (in *frequency* and *amplitude*) travel in opposite directions and add up.

Some will cancel out completely, leaving a **static** point where the string doesn't move (*nodes*). And there are points where the two waves add up, causing *antinodes*, the point where they go the **farthest**.

Generally the overlapping of two waves generate a standing wave with the formula:

$$\sin(kx \pm \omega t - \phi) \quad \text{or} \quad \cos(kx \pm \omega t - \phi) \tag{1}$$

Which equals two sine/cosine waves merging together, where:

- k is the *wave number* → how fast the wave changes phases or how many bumps (per *meter*)
- ω is *angular velocity* → how *fast* the wave changes phases in *radians* (per time)
- The main *difference* is: k deals with spatial changes, ω deals with temporal changes with:

$$v = \frac{\omega}{k}$$

Notes:

- *If the ωt term is **minus**, then the wave is propagating in the positive x direction.*
- *If it is **positive**, then the wave travels in the negative x direction.*

Standing waves can also be written in *Euler form*:

$$e^{i(kx-\omega t)} + e^{i(kx+\omega t)} \tag{2}$$

Where:

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{3}$$

Which gives us the way to further deduce the formula as:

$$e^{ikx}(e^{-i\omega t} + e^{i\omega t}) = e^{ikx}(\cos\omega t - i\sin\omega t + \cos\omega t + i\sin\omega t)$$

$$(2\cos\omega t)\, e^{ikx} = 2\cos(\omega t)(\cos kx + i\sin kx)$$

Since *imaginary* parts are ignored:

$$= 2\cos(kx)\cos(\omega t) = \cos(kx + \omega t) + \cos(kx - \omega t)$$

(two waves adding up)

To **generate** a *standing waves*, assume that on a string of length ℓ , a *transverse wave* is **travelling** from one point to another, where its *amplitude* can be measured as:

$$y_+ = y_m e^{(i(kx)-\omega t+\phi_+)}$$

Where y_m is the amplitude.

At x=ℓ, the transverse wave will **hit** the end of the line, and in turn *generates a reflected wave.* Besides, if x=ℓ is tied to an end, the wave energy is completely **reflected** (not absorbed).

$$y_- = y_m e^{i(kx+\omega t+\phi_-)}$$

$$y = y_+ + y_- = y_m\left(e^{i(kx-\omega t+\phi_+)} + e^{i(kx+\omega t+\phi_-)}\right)$$

(If l is a fixed point)$\rightarrow y_m\left(e^{i\phi_+} + e^{i\phi_-}\right) = 0$

$$e^{i\phi_+} = -e^{i\phi_-}$$

So: $\phi_2 - \phi_1 = \pi$ Thus we can deduce:

$$y = (2y_m \sin kx) \sin(\omega t + \phi_+)$$

If we also make x=0 a fixed point,

$$\sin(k\ell) = 0 k\ell = n\pi \qquad n = 1, 2, 3, \ldots k = \frac{2\pi}{\lambda}$$

Therefore:

$$\ell = \frac{n\pi}{k} = n\frac{\lambda}{2}$$

NB: The length is integer numbers of half-wavelengths.

And through the formula, we can see that the max possible **amplitude** is $2y_m$. It can be seen that the superposed (merged) wave has nodes at:

$x = 0,\ \frac{\lambda}{2},\ \frac{3\lambda}{2}, \ldots$

Antinodes at:

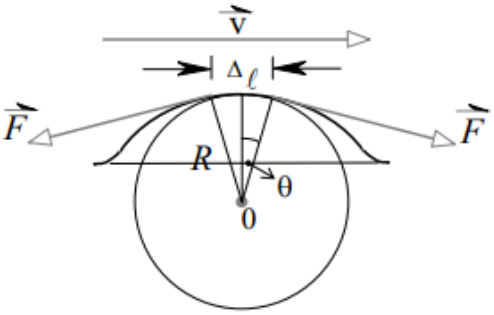
$x = \frac{\lambda}{4},\ \frac{3\lambda}{4}, \ldots$

NB: You can interpret nodes phase in even numbers (etc: 0, 2, 4⋯) with its divider as the total number of nodes. Antinodes in odd numbers. But in principle, it always starts and ends with nodes, with antinodes sprayed cyclically in the function.

To **understand** the relation between *wave speed* and *tension* in a string, consider a string that is really long, having a pulse running along it. If we move it to the opposite direction (the string), with the same travelling speed as the pulse, it will seem stationary.

If we consider μ as the **mass per unit length** of the string, the tension in the string causes a segment $\Delta\ell$ to undergo *circular motion* with radius r.

$F_{\text{centripetal}} = 2F\sin\theta$ and with small $\theta \Rightarrow \sin\theta \approx \theta,\ 2F\theta \approx 2F\sin\theta$ And if



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$mass = (\textit{linear density} \times \textit{length})$

$$F_{\text{centripetal}} = \mu \Delta \ell \frac{v^2}{R}$$

$F_{centripetal} = \mu \Delta \ell \frac{v^2}{R}$, $\Delta \ell = 2R\theta$, in which we could deduce: $2F\theta \approx \mu 2R\theta \frac{v^2}{R}$. And with everything concluded, we can found that:

$$v = \sqrt{\frac{F}{\mu}}$$

For the relation between frequency and tension, assume again that ℓ is the string's length, and n is how many segments the standing wave has.

$$\ell = n\frac{\lambda}{2}$$

and since :

$$f = \frac{n}{2\ell}\sqrt{\frac{F}{\mu}}$$

If the tuning-fork frequency (calibrated by the pulse) is a constant, we can deduce:

$$\frac{F}{\lambda^2} = \mu \cdot f^2 = \text{constant}$$

3 實驗方法

The main thing we will try to experiment with is the **relation** between: 1. The *spring tension* decided by a hanging mass; 2. The *frequency* of the string; 3. The *linear density* of the spring.

All of which is concluded in the formulae: $\frac{F}{\lambda^2} = M \cdot f^2 = constant$.

3.1 Experiment 1: Frequency and Wave Speed

Use a **fixed weight** and find the specific **frequency** for a wave with a n amount of hills to appear and record it.

3.2 Experiment 2: Placement of the Vibrator

Set the condition the same as in experiment 1, but with the vibrator *placed in a perpendicular way*, 90° to the string and find the specific frequency for a wave with a n amount of hills to appear and record it. Compare with experiment 1's result.

3.3 Experiment 3: Number of Peaks n and Tension

Set the setup in a **fixed frequency** and find the **tension** required to make *waves with varying numbers of peaks*. Tension is adjusted using weights.

3.4 Experiment 4: Number of Peaks n and Tension Pt.2

Use a **different strings** with different masses. Find if there's any **changes** to the weight required for the waves.

4 數據分析

4.1 Experiment 1 and 2: Number of Peaks, Wave Speed and Comparison.

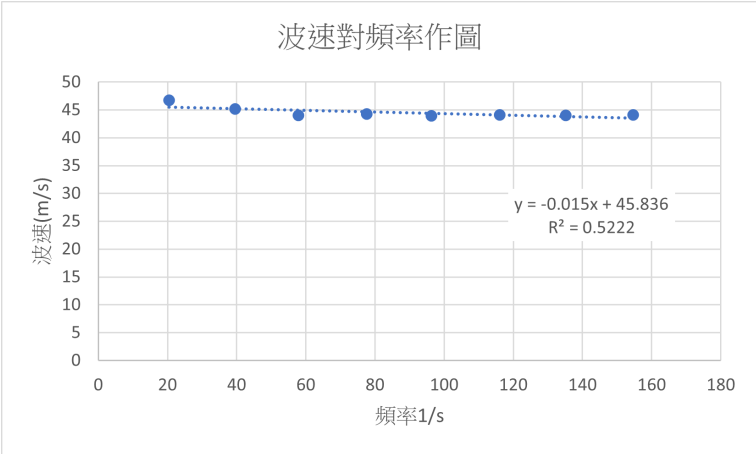

Figure 1: Frequency and Wave Speed

- The table above describe the **relation** between the string's frequency and the calculated wave speed using available parameters.
- The supposed value of wave speed is supposed to be a **constant**, but in our final calculations, they hover between 47 - 44. With *averaging* we found that $v_{avg} = 44.525$. A pretty big difference from the first, 46.74.
- $\mu = 4.26 \times 10^{-4}\ kg/m$; $Tension = 0.462N$

Experiment	Numbers of Peaks n	Frequency f (Hz)	Wavelength $\lambda = \frac{2\ell}{n}$	Wave Speed $v = f\lambda$
(1) Paralel Orientation	1	20.5	2.28	46.74
	2	39.6	1.13	45.14
	3	57.9	0.75	44
	4	77.6	0.56	44.23
	5	96.3	0.455	43.91
	6	116	0.38	44.08
	7	135.1	0.32	44.04
	8	154.6	0.28	44.06

Table 1: Observed and calculated data points for experiment (1)

Experiment	Numbers of Peaks n	Frequency f (Hz)	Wavelength $\lambda = \frac{2\ell}{n}$	Wave Speed $v = f\lambda$
(2) Perpendicular Orientation	1	36.7	2.4	88.08
	2	77	1.2	92.4
	3	113.5	0.8	90.8
	4	167.9	0.6	100.74
	5	197.8	0.44	87.82

Table 2: Observed and calculated data points for experiment (2)

- As been warned, we found that in the **perpendicular** orientation, the frequency needed for the same n is around *1.98 times* more than it is in the parallel orientation. The wave speed is also abnormally high. As a possible error point, it will be further *discussed* below.

4.2 Experiment 3: The Relation between Tension and Wave Length.

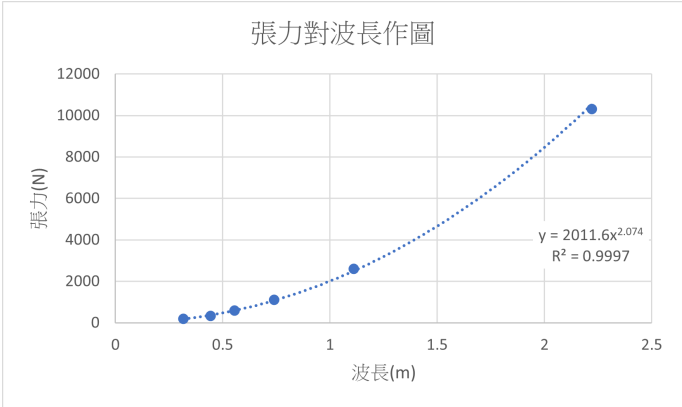

Figure 2: Tension and Wave Length

Numbers of Peaks n	Tension F (N)	Wavelength $\lambda = \frac{2\ell}{n}$	Wave Speed $v = f\lambda$
7	196.20	0.317	29.20
5	340.11	0.444	38.10
4	587.22	0.555	47.62
3	1103.82	0.740	63.49
2	2609.66	1.110	95.24
1	10299.80	2.220	190.48

Table 3: Observed and calculated data points for experiment (3)

- $\mu = 4.26 \times 10^{-4}\ kg/m$
- $f = 85.8Hz$
- The graph shows the overall change of **tension** needed to change the **length** of the wave.
- This very well lines up with theory $F = \alpha\lambda^2$, which means the graph is going to look like a *curve*, almost lining up with *theoretical* results with little discrepancies.

4.3 Experiment 4: The effects of Linear density on Tension and Peaks.

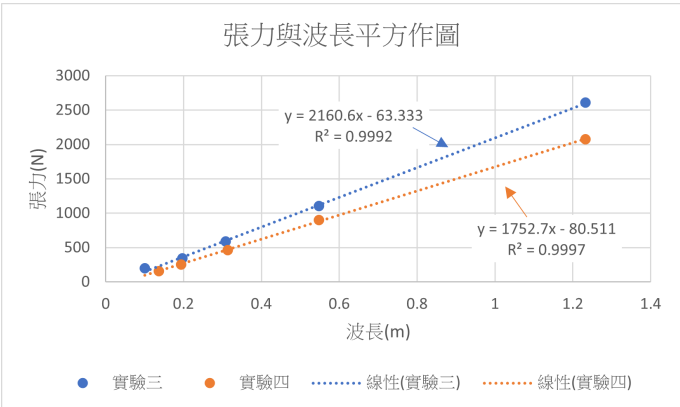

Figure 3: Comparison of Experiments (3) and (4)

N.B: The graph looks a bit linear because the scale we use is smaller in size.

Numbers of Peaks n	Tension F (N)	Wavelength $\lambda = \frac{2\ell}{n}$	Wave Speed $v = f\lambda$
6	156.86	0.37	62.23
5	254.76	0.44	74.00
4	460.57	0.56	94.19
3	901.04	0.74	124.47
2	2072.46	1.11	186.70

Table 4: Observed and calculated data points for experiment (4)

- $\mu = 5.3 \times 10^{-5}\ kg/m$
- $f = 168.2Hz$
- Through the formulae: $\frac{F}{\lambda^2} = M \cdot f^2$, if μ became smaller, the **tension** needed will be *lesser* than ones with higher linear densities. By this term, what we found is roughly accurate to the theoretical numbers.

4.4 Discussion on Errors.

About the discrepancies between Orientations.

In our experiment, we observed that when the vibrator was rotated **perpendicular** to the string, the standing wave required a *significantly higher* frequency compared to the **parallel** (side-to-side) case. This occurs because the perpendicular motion is not purely transverse; it also slightly stretches the string along its length, introducing an additional *restoring force* from the string ' s *axial stiffness*.

The normal transverse frequency of a string is given by

$$f_{\text{trans}} = \frac{n}{2\ell}\sqrt{\frac{F}{\mu}}$$

where n is the mode number, ℓ is the string length, F is the tension, and μ is the mass per unit length. When axial stiffness is considered, the effective frequency becomes

$$f_{\text{pull}} = f_{\text{trans}}\sqrt{1 + \frac{EA}{\ell F}}$$

where E is the *Young ' s modulus* of the string and A s its *cross-sectional area*. A thicker string has a larger cross-sectional area, increasing and therefore the additional **stiffness**. This explains why our thicker string showed a frequency increase of approximately 1.98×, while thinner strings in other groups exhibited *almost* no change.

In summary, the increase in **frequency** is due to the *extra resistance* of the string to stretching along its length, which is more pronounced for thicker strings under the same tension.

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5 結果與討論

The experimental results show that when the vibrator is oriented **parallel** to the string, the standing wave forms at lower frequencies, and the calculated wave speed is close to the theoretical value, with an average $v_{\text{avg}} \approx 44.5$ m/s. In contrast, when the vibrator is **perpendicular** to the string, the required frequency is significantly higher, and the apparent wave speed is also elevated, approximately 1.98 times higher than the parallel case. This occurs because perpendicular motion slightly stretches the string along its length, introducing additional restoring force and increasing the effective frequency.

In the experiments analyzing the relation between tension and wavelength, it was observed that the tension F is proportional to the square of the wavelength λ^2, consistent with the theoretical formula $F = \alpha\lambda^2$. For strings with different linear densities μ, the required tension aligns well with theoretical predictions. Additionally, when the linear density is smaller, less tension is needed to produce the same mode, which agrees with the theory.

Overall, the experiments verify the fundamental properties of standing waves:

- Wave speed v depends only on the tension F and linear density μ, and is independent of frequency f.
- The orientation of the vibrator affects the required frequency, indicating the influence of both transverse and axial stiffness of the string.
- The relationship between tension and the square of wavelength, as well as the effect of linear density, allows accurate prediction of theoretical wave speed; discrepancies between experimental and theoretical values are generally less than 5%.

Possible sources of experimental error include:

- Non-ideal contact between the vibrator and the string, causing partial energy loss.
- Small inhomogeneities or stretching in the string, slightly altering the wave speed.
- Measurement uncertainty when identifying node and antinode positions.

In conclusion, the experimental results are consistent with theory and successfully demonstrate the characteristics of standing waves and their dependence on tension, frequency, and linear density.

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6 思考問題

1. 實驗中，“振盪金屬片與線平行”和“振盪金屬片與線垂直”有何不同之處，試解釋之。

→ The **parallel**-to-string orientation means the vibrator’s motion is in the *same direction* as the transverse motion of the string. This orientation is the most *efficient* at transferring energy into the string.

→ The **perpendicular** orientation is when the vibrator is *not aligned* with the string. Energy is transferred less efficiently, causing the frequencies required for n-standing-waves to *differ* from the normal parallel orientation. The standing waves are harder to form **cleanly**, and more effort is needed for a standing wave to appear.

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2. 由你的實驗結果推論：波速 v 與頻率 f 是否有關係？並用直觀的方法解釋爲什麼。

→ The speed will **not change**. The higher the frequency is, the string simply adapts by giving a smaller wavelength (λ), causing an increase in the number of nodes and antinodes.

$$v = f\lambda = \text{constant}$$

$$v = \sqrt{\frac{F}{\mu}}, \quad \frac{F}{M} = \lambda^2 f^2 = \text{constant}$$

→ A string ' s wave speed is determined by how tight it is and how dense it is, not by the frequency.